EXPERIMENT 4

AIM: Install ReactJS and Environment Setup.

SOLUTION:

- ✓ React is a JavaScript library used for building user interfaces and single-page applications. To use React, we need Node.js to manage packages and run the development server.

Node.js provides:

- ✓ npm (Node Package Manager)
- ✓ Development server support
- ✓ Build tools

Modern React projects are created using tools:

1. Vite

2. Create React App

Procedure / Solution

Step 1: Install Node.js

Go to official website: https://nodejs.org

Download LTS (Long Term Support) version: Node.js V20.16.0(LTS)

Install with default settings

Verify Installation

Open Command Prompt and type:

>node -v

v20.16.0

>npm -v

10.8.1

Step 2: Install VS Code

Download from https://code.visualstudio.com

Install normally

Open VS Code

Step 3: Create React Application Using Vite

Open Command Prompt and type:

>npm create vite@latest myapp

Select:

Framework → React

Variant → JavaScript

Then run:

cd myapp

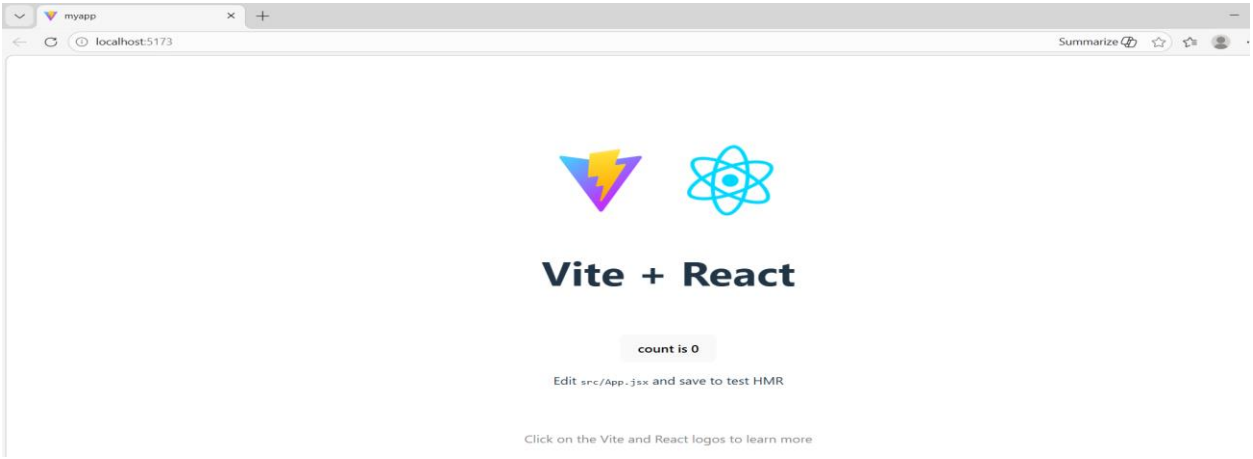
npm run dev

Step 4: Run Application

After running npm run dev, you will see:

http://localhost:5173/

Open this URL in browser.



Or

Step 3: Create React Application

Open Command Prompt and Type:

>npx create-react-app react-lab_old

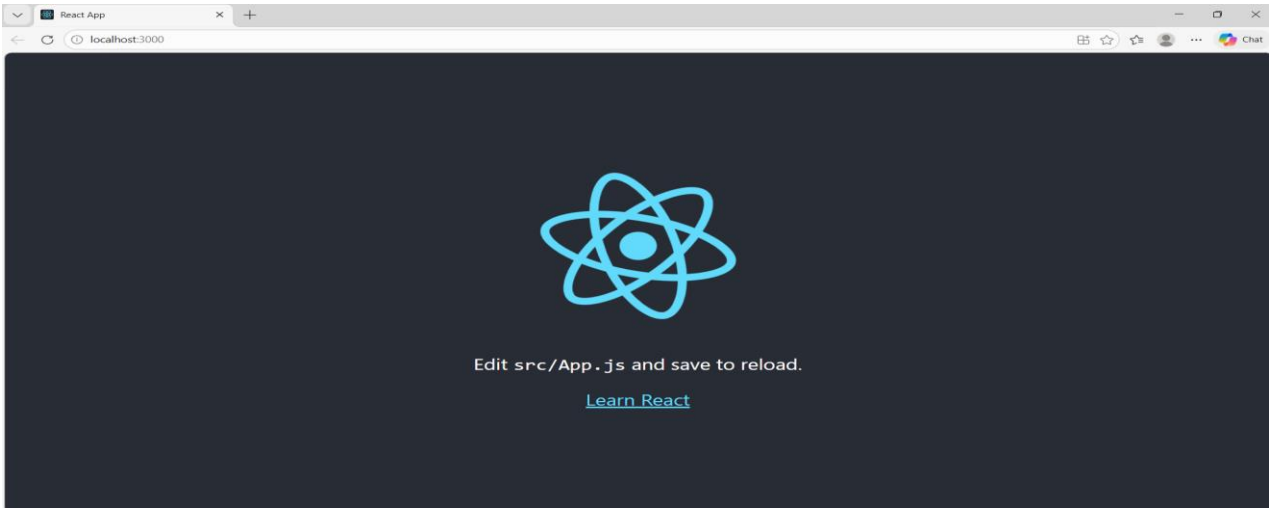
Move into project folder:

>cd react-lab

Step 4: Start React Application

>npm start

The application runs at:

http://localhost:3000

Output: